restart

Solution with two media

$$eq1 := \frac{(epsilonEff - 1)}{epsilonEff + 2} = \frac{(epsilon1 - 1)}{epsilon1 + 2} \cdot eta1 + \frac{(epsilon2 - 1)}{epsilon2 + 2} \cdot eta2$$

$$\frac{epsilonEff - 1}{epsilonEff + 2} = \frac{(\varepsilon 1 - 1)\ \eta 1}{\varepsilon 1 + 2} + \frac{(\varepsilon 2 - 1)\ \eta 2}{\varepsilon 2 + 2} \quad \textbf{(1.1)}$$

$eta1 := (1e0 - LiquidFraction);\ eta2 := LiquidFraction;$

$$1. - LiquidFraction$$

$$LiquidFraction \quad \textbf{(1.2)}$$

Analytical solution

$epsilonEff := epsilonEffR + I \cdot epsilonEffC;\ epsilon1 := epsilon1R + I \cdot epsilon1C;\ epsilon2 := epsilon2R + I \cdot epsilon2C;$

$$epsilonEffR + \mathrm{I}\ epsilonEffC$$

$$epsilon1R + \mathrm{I}\ epsilon1C$$

$$epsilon2R + \mathrm{I}\ epsilon2C \quad \textbf{(1.3)}$$

$eq2 := evalc(eq1)$

$$\frac{(epsilonEffR - 1)\ (epsilonEffR + 2)}{(epsilonEffR + 2)^2 + epsilonEffC^2} + \frac{epsilonEffC^2}{(epsilonEffR + 2)^2 + epsilonEffC^2} + \mathrm{I}\left(\frac{epsilonEffC\ (epsilonEffR + 2)}{(epsilonEffR + 2)^2 + epsilonEffC^2} - \frac{(epsilonEffR - 1)\ epsilonEffC}{(epsilonEffR + 2)^2 + epsilonEffC^2}\right) = \left(\frac{(epsilon1R - 1)\ (epsilon1R + 2)}{(epsilon1R + 2)^2 + epsilon1C^2} + \frac{epsilon1C^2}{(epsilon1R + 2)^2 + epsilon1C^2}\right)(1. - LiquidFraction) + \left(\frac{(epsilon2R - 1)\ (epsilon2R + 2)}{(epsilon2R + 2)^2 + epsilon2C^2} + \frac{epsilon2C^2}{(epsilon2R + 2)^2 + epsilon2C^2}\right) LiquidFraction + \mathrm{I}\left(\left(\frac{epsilon1C\ (epsilon1R + 2)}{(epsilon1R + 2)^2 + epsilon1C^2} - \frac{(epsilon1R - 1)\ epsilon1C}{(epsilon1R + 2)^2 + epsilon1C^2}\right)(1. - LiquidFraction) + \left(\frac{epsilon2C\ (epsilon2R + 2)}{(epsilon2R + 2)^2 + epsilon2C^2} - \frac{(epsilon2R - 1)\ epsilon2C}{(epsilon2R + 2)^2 + epsilon2C^2}\right) LiquidFraction\right) \quad \textbf{(1.4)}$$

Two equations can be built from this

$$eq2r := \frac{(epsilonEffR-1)(epsilonEffR+2)}{(epsilonEffR+2)^2+epsilonEffC^2}+\frac{epsilonEffC^2}{(epsilonEffR+2)^2+epsilonEffC^2} = \left(\frac{(epsilon1R-1)(epsilon1R+2)}{(epsilon1R+2)^2+epsilon1C^2}+\frac{epsilon1C^2}{(epsilon1R+2)^2+epsilon1C^2}\right)(1. - LiquidFraction) + \left(\frac{(epsilon2R-1)(epsilon2R+2)}{(epsilon2R+2)^2+epsilon2C^2} + \frac{epsilon2C^2}{(epsilon2R+2)^2+epsilon2C^2}\right) LiquidFraction$$

$$\frac{(epsilonEffR-1)(epsilonEffR+2)}{(epsilonEffR+2)^2+epsilonEffC^2}+\frac{epsilonEffC^2}{(epsilonEffR+2)^2+epsilonEffC^2} = \left(\frac{(epsilon1R-1)(epsilon1R+2)}{(epsilon1R+2)^2+epsilon1C^2}+\frac{epsilon1C^2}{(epsilon1R+2)^2+epsilon1C^2}\right)(1. - LiquidFraction) + \left(\frac{(epsilon2R-1)(epsilon2R+2)}{(epsilon2R+2)^2+epsilon2C^2} + \frac{epsilon2C^2}{(epsilon2R+2)^2+epsilon2C^2}\right) LiquidFraction \tag{1.5}$$

$$eq2c := \frac{epsilonEffC\,(epsilonEffR+2)}{(epsilonEffR+2)^2+epsilonEffC^2}-\frac{(epsilonEffR-1)\,epsilonEffC}{(epsilonEffR+2)^2+epsilonEffC^2} = \left(\frac{epsilon1C\,(epsilon1R+2)}{(epsilon1R+2)^2+epsilon1C^2}-\frac{(epsilon1R-1)\,epsilon1C}{(epsilon1R+2)^2+epsilon1C^2}\right)(1. - LiquidFraction) + \left(\frac{epsilon2C\,(epsilon2R+2)}{(epsilon2R+2)^2+epsilon2C^2} - \frac{(epsilon2R-1)\,epsilon2C}{(epsilon2R+2)^2+epsilon2C^2}\right) LiquidFraction$$

$$\frac{epsilonEffC\,(epsilonEffR+2)}{(epsilonEffR+2)^2+epsilonEffC^2}-\frac{(epsilonEffR-1)\,epsilonEffC}{(epsilonEffR+2)^2+epsilonEffC^2} = \left(\frac{epsilon1C\,(epsilon1R+2)}{(epsilon1R+2)^2+epsilon1C^2}-\frac{(epsilon1R-1)\,epsilon1C}{(epsilon1R+2)^2+epsilon1C^2}\right)(1. - LiquidFraction) + \left(\frac{epsilon2C\,(epsilon2R+2)}{(epsilon2R+2)^2+epsilon2C^2} - \frac{(epsilon2R-1)\,epsilon2C}{(epsilon2R+2)^2+epsilon2C^2}\right) LiquidFraction \tag{1.6}$$

So we can find analytical solutions for both

$sol := solve(\{eq2r, eq2c\}, \{epsilonEffR, epsilonEffC\})$

$$\Big\{ epsilonEffC = \big(-4.\ epsilon1C\ LiquidFraction\ epsilon2R \tag{1.7}$$

$$+ 4.\ epsilon1C\ epsilon2R + epsilon1C\ epsilon2C^2 + epsilon1C\ epsilon2R^2$$

$+ 4.\ epsilon2C\ LiquidFraction\ epsilon1R$

$+ epsilon2C\ LiquidFraction\ epsilon1C^2 - 4.\ epsilon1C\ LiquidFraction$

$+ epsilon2C\ LiquidFraction\ epsilon1R^2 + 4.\ epsilon1C$

$- 1.\ epsilon1C\ LiquidFraction\ epsilon2C^2 + 4.\ epsilon2C\ LiquidFraction$

$- 1.\ epsilon1C\ LiquidFraction\ epsilon2R^2) \Big/$

$(2.\ epsilon1R\ LiquidFraction\ epsilon2R + 4. - 2.\ LiquidFraction\ epsilon2R^2$

$- 4.\ LiquidFraction\ epsilon2R - 2.\ LiquidFraction\ epsilon2C^2$

$+ 4.\ epsilon1R\ LiquidFraction + epsilon2R^2 + 4.\ epsilon2R + epsilon2C^2$

$+ 2.\ epsilon1C\ epsilon2C\ LiquidFraction + LiquidFraction^2\ epsilon2R^2$

$+ epsilon1R^2\ LiquidFraction^2 - 2.\ LiquidFraction^2\ epsilon2R\ epsilon1R$

$- 2.\ epsilon2C\ LiquidFraction^2\ epsilon1C + epsilon1C^2\ LiquidFraction^2$

$+ epsilon2C^2\ LiquidFraction^2),\ epsilonEffR =$

$-(1.\ (2.\ epsilon1C^2\ LiquidFraction^2 - 2.\ LiquidFraction\ epsilon2R^2$
$- 4.\ epsilon1R - 4.\ epsilon1R\ epsilon2R - 1.\ epsilon1R\ epsilon2C^2$
$+ 4.\ epsilon1R\ LiquidFraction\ epsilon2R - 4.\ LiquidFraction\ epsilon2R$
$- 1.\ epsilon1R^2\ LiquidFraction\ epsilon2R$
$+ epsilon1R\ LiquidFraction\ epsilon2R^2 - 2.\ epsilon1R^2\ LiquidFraction$
$+ epsilon1R\ LiquidFraction\ epsilon2C^2$
$- 1.\ epsilon1C^2\ LiquidFraction\ epsilon2R + 2.\ epsilon2C^2\ LiquidFraction^2$
$- 4.\ epsilon2C\ LiquidFraction^2\ epsilon1C + 4.\ epsilon1R\ LiquidFraction$
$- 1.\ epsilon1R\ epsilon2R^2 + 2.\ epsilon1R^2\ LiquidFraction^2$
$- 4.\ LiquidFraction^2\ epsilon2R\ epsilon1R - 2.\ LiquidFraction\ epsilon2C^2$
$- 2.\ epsilon1C^2\ LiquidFraction + 4.\ epsilon1C\ epsilon2C\ LiquidFraction$

$$+ 2.\, LiquidFraction^2\, epsilon2R^2))\Big/(2.\, epsilon1R\, LiquidFraction\, epsilon2R + 4. - 2.\, LiquidFraction\, epsilon2R^2 - 4.\, LiquidFraction\, epsilon2R - 2.\, LiquidFraction\, epsilon2C^2 + 4.\, epsilon1R\, LiquidFraction + epsilon2R^2 + 4.\, epsilon2R + epsilon2C^2 + 2.\, epsilon1C\, epsilon2C\, LiquidFraction + LiquidFraction^2\, epsilon2R^2 + epsilon1R^2\, LiquidFraction^2 - 2.\, LiquidFraction^2\, epsilon2R\, epsilon1R - 2.\, epsilon2C\, LiquidFraction^2\, epsilon1C + epsilon1C^2\, LiquidFraction^2 + epsilon2C^2\, LiquidFraction^2)\}$$

$\#with(codegen)$

$\#espilonEffR := sol[2]; \#cost(espilonEffR);$

Define a function for optimizing the beast.

$$\#epsilonEffRe := (epsilon1R, epsilon1C, epsilon2R, epsilon2C, LiquidFraction) \to -(1.\,(2.\, epsilon1C^2\, LiquidFraction^2 - 2.\, LiquidFraction\, epsilon2R^2 - 4.\, epsilon1R + epsilon1R\, LiquidFraction\, epsilon2C^2 - 1.\, epsilon1C^2\, LiquidFraction\, epsilon2R + 4.\, epsilon1R\, LiquidFraction - 1.\, epsilon1R^2\, LiquidFraction\, epsilon2R + epsilon1R\, LiquidFraction\, epsilon2R^2 + 4.\, epsilon1R\, LiquidFraction\, epsilon2R - 1.\, epsilon1R\, epsilon2C^2 - 2.\, epsilon1C^2\, LiquidFraction + 4.\, epsilon1C\, epsilon2C\, LiquidFraction + 2.\, epsilon2C^2\, LiquidFraction^2 - 2.\, epsilon1R^2\, LiquidFraction - 1.\, epsilon1R\, epsilon2R^2 - 4.\, epsilon1R\, epsilon2R + 2.\, epsilon1R^2\, LiquidFraction^2 - 4.\, LiquidFraction\, epsilon2R - 2.\, LiquidFraction\, epsilon2C^2 - 4.\, epsilon2C\, LiquidFraction^2\, epsilon1C + 2.\, LiquidFraction^2\, epsilon2R^2 - 4.\, LiquidFraction^2\, epsilon2R\, epsilon1R))\Big/(2.\, epsilon1R\, LiquidFraction\, epsilon2R + 4. + 4.\, epsilon1R\, LiquidFraction - 2.\, LiquidFraction\, epsilon2R^2 - 4.\, LiquidFraction\, epsilon2R - 2.\, LiquidFraction\, epsilon2C^2 + epsilon2R^2 + 4.\, epsilon2R + epsilon2C^2 + 2.\, epsilon1C\, epsilon2C\, LiquidFraction + epsilon1R^2\, LiquidFraction^2 + LiquidFraction^2\, epsilon2R^2 - 2.\, LiquidFraction^2\, epsilon2R\, epsilon1R - 2.\, epsilon2C\, LiquidFraction^2\, epsilon1C + epsilon2C^2\, LiquidFraction^2 + epsilon1C^2\, LiquidFraction^2)$$

$\#optimize(epsilonEffRe); cost(optimize(epsilonEffRe))$

$\#espilonEffImag := sol[1]; \#cost(espilonEffImag);$

$$\#epsilonEffImag := (epsilon1R, epsilon1C, epsilon2R, epsilon2C, LiquidFraction) \to (4.\, epsilon2C\, LiquidFraction - 1.\, epsilon1C\, LiquidFraction\, epsilon2C^2 + 4.\, epsilon1C\, epsi + 4.\, epsilon2C\, LiquidFraction\, epsilon1R + epsilon2C\, LiquidFraction\, epsilon1C^2 - 4.\, epsilon1C\, LiquidFraction + epsilon2C\, LiquidFraction\, epsilon1R^2 + 4.\, epsilon1C + epsilon1C\, epsilon2C^2 + epsilon1C\, epsilon2R^2 - 1.\, epsilon1C\, LiquidFraction\, epsilon2R^2)\Big/(2.\, epsilon1R\, LiquidFraction\, epsilon2R + 4. + 4.\, epsilon1R\, LiquidFraction - 2.\, LiquidFraction\, epsilon2R^2 - 4.\, LiquidFraction\, epsilon2R - 2.\, LiquidFraction\, epsilon2C^2 + epsilon2R^2 + 4.\, epsilon2R + epsilon2C^2 + 2.\, epsilon1C\, epsilon2C\, LiquidFraction + epsilon1R^2\, LiquidFraction^2 + LiquidFraction^2\, epsilon2R^2$$

$- 2.\ LiquidFraction^2\ epsilon2R\ epsilon1R - 2.\ epsilon2C\ LiquidFraction^2\ epsilon1C + epsilon2C^2\ LiquidFraction^2 + epsilon1C^2\ LiquidFraction^2)$

$\#cost(espilonEffImag);\ cost(optimize(epsilonEffImag))$ **Optimization would not be huge.**

VERIFICATIONS

$epsilonEffRe := sol[2] : epsilonEffImag := sol[1] :$

$eval(epsilonEffRe, LiquidFraction = 0);\ eval(epsilonEffImag, LiquidFraction = 0)$

$$epsilonEffR = -(1.\ (-4.\ epsilon1R - 4.\ epsilon1R\ epsilon2R - 1.\ epsilon1R\ epsilon2C^2 - 1.\ epsilon1R\ epsilon2R^2))\Big/(4. + epsilon2R^2 + 4.\ epsilon2R + epsilon2C^2)$$

$$epsilonEffC = (4.\ epsilon1C\ epsilon2R + epsilon1C\ epsilon2C^2 + epsilon1C\ epsilon2R^2 + 4.\ epsilon1C)\Big/(4. + epsilon2R^2 + 4.\ epsilon2R + epsilon2C^2) \tag{1.8}$$

Numerical application using the found analytical formula

$epsilonSi := 13.64 + I\cdot 48$

$$13.64 + 48.\ \mathrm{I} \tag{1.9}$$

$epsilonLSi := -17 + I\cdot 49$

$$-17 + 49\ \mathrm{I} \tag{1.10}$$

$epsilon1R := \mathrm{Re}(epsilonSi);\ epsilon1C := \mathrm{Im}(epsilonSi);\ epsilon2R := \mathrm{Re}(epsilonLSi);\ epsilon2C := \mathrm{Im}(epsilonLSi)$

$$13.64$$

$$48.$$

$$-17$$

$$49 \tag{1.11}$$

$eval(epsilonEffRe, LiquidFraction = 0.5);\ eval(epsilonEffImag, LiquidFraction = 0.5)$

$$epsilonEffR = -1.396030969$$

$$epsilonEffC = 53.33594371 \tag{1.12}$$

Numerical application using Maple solver, without splitting Re and Im.

$\varepsilon_1 := epsilonSi;\ \varepsilon_2 := epsilonLSi$

$$13.64 + 48.\ \mathrm{I}$$

$$-17 + 49\ \mathrm{I} \tag{1.13}$$

$eval\left(solve\left(eq1, \varepsilon_{eff}\right), LiquidFraction = 0.5\right)$

$$-1.396030950 + 53.33594372\,\mathrm{I} \tag{1.14}$$

OPTIMIZE the writing of analytical formula

$with(codegen, optimize, makeproc, cost, prep2trans)$

$$[optimize, makeproc, cost, prep2trans] \tag{1.15}$$

$$\begin{aligned}
epsilonEffR :=\, & \big(4.\, LiquidFraction^2\, \varepsilon_{r2} \varepsilon_{r1} - 2.\, \varepsilon_{r1}^2\, LiquidFraction^2 \\
& - 4.\, \varepsilon_{c2}\, LiquidFraction\, \varepsilon_{c1} + \varepsilon_{r1}\, \varepsilon_{c2}^2 - 4.\, \varepsilon_{r1}\, LiquidFraction\, \varepsilon_{r2} \\
& + 4.\, \varepsilon_{c2}\, LiquidFraction^2\, \varepsilon_{c1} - 2.\, \varepsilon_{c1}^2\, LiquidFraction^2 + 2.\, \varepsilon_{r1}^2\, LiquidFraction \\
& + 2.\, LiquidFraction\, \varepsilon_{r2}^2 - 2.\, \varepsilon_{c2}^2\, LiquidFraction^2 + 4.\, \varepsilon_{r1} + 4.\, \varepsilon_{r1}\, \varepsilon_{r2} \\
& - 4.\, \varepsilon_{r1}\, LiquidFraction - 2.\, LiquidFraction^2\, \varepsilon_{r2}^2 + \varepsilon_{r1}\, \varepsilon_{r2}^2 + 2.\, LiquidFraction\, \varepsilon_{c2}^2 + \\
& \varepsilon_{r1}^2\, LiquidFraction\, \varepsilon_{r2} - 1.\, \varepsilon_{r1}\, LiquidFraction\, \varepsilon_{r2}^2 + 4.\, LiquidFraction\, \varepsilon_{r2} \\
& - 1.\, \varepsilon_{r1}\, LiquidFraction\, \varepsilon_{c2}^2 + \varepsilon_{c1}^2\, LiquidFraction\, \varepsilon_{r2} + 2.\, \varepsilon_{c1}^2\, LiquidFraction\big) \Big/ \\
& \big(LiquidFraction^2\, \varepsilon_{r2}^2 + \varepsilon_{r1}^2\, LiquidFraction^2 - 2.\, LiquidFraction^2\, \varepsilon_{r2} \varepsilon_{r1} + \\
& \varepsilon_{c2}^2\, LiquidFraction^2 + \varepsilon_{c1}^2\, LiquidFraction^2 + 2.\, \varepsilon_{c2}\, LiquidFraction\, \varepsilon_{c1} \\
& - 2.\, \varepsilon_{c2}\, LiquidFraction^2\, \varepsilon_{c1} + \varepsilon_{r2}^2 + 4.\, \varepsilon_{r2} + \varepsilon_{c2}^2 + 4. - 2.\, LiquidFraction\, \varepsilon_{r2}^2 \\
& - 4.\, LiquidFraction\, \varepsilon_{r2} - 2.\, LiquidFraction\, \varepsilon_{c2}^2 + 4.\, \varepsilon_{r1}\, LiquidFraction \\
& + 2.\, \varepsilon_{r1}\, LiquidFraction\, \varepsilon_{r2}\big)
\end{aligned}$$

$$\begin{aligned}
& \big(4.\, LiquidFraction^2\, \varepsilon_{r2} \varepsilon_{r1} - 2.\, \varepsilon_{r1}^2\, LiquidFraction^2 - 4.\, \varepsilon_{c2}\, LiquidFraction\, \varepsilon_{c1} \\
& + \varepsilon_{r1}\, \varepsilon_{c2}^2 - 4.\, \varepsilon_{r1}\, LiquidFraction\, \varepsilon_{r2} + 4.\, \varepsilon_{c2}\, LiquidFraction^2\, \varepsilon_{c1} - 2. \\
& \varepsilon_{c1}^2\, LiquidFraction^2 + 2.\, \varepsilon_{r1}^2\, LiquidFraction + 2.\, LiquidFraction\, \varepsilon_{r2}^2 - 2. \\
& \varepsilon_{c2}^2\, LiquidFraction^2 + 4.\, \varepsilon_{r1} + 4.\, \varepsilon_{r1}\, \varepsilon_{r2} - 4.\, \varepsilon_{r1}\, LiquidFraction \\
& - 2.\, LiquidFraction^2\, \varepsilon_{r2}^2 + \varepsilon_{r1}\, \varepsilon_{r2}^2 + 2.\, LiquidFraction\, \varepsilon_{c2}^2 + \\
& \varepsilon_{r1}^2\, LiquidFraction\, \varepsilon_{r2} - 1.\, \varepsilon_{r1}\, LiquidFraction\, \varepsilon_{r2}^2 + 4.\, LiquidFraction\, \varepsilon_{r2} \\
& - 1.\, \varepsilon_{r1}\, LiquidFraction\, \varepsilon_{c2}^2 + \varepsilon_{c1}^2\, LiquidFraction\, \varepsilon_{r2} + 2.\, \varepsilon_{c1}^2\, LiquidFraction\big) \Big/ \\
& \big(LiquidFraction^2\, \varepsilon_{r2}^2 + \varepsilon_{r1}^2\, LiquidFraction^2 - 2.\, LiquidFraction^2\, \varepsilon_{r2} \varepsilon_{r1} + \\
& \varepsilon_{c2}^2\, LiquidFraction^2 + \varepsilon_{c1}^2\, LiquidFraction^2 + 2.\, \varepsilon_{c2}\, LiquidFraction\, \varepsilon_{c1} \\
& - 2.\, \varepsilon_{c2}\, LiquidFraction^2\, \varepsilon_{c1} + \varepsilon_{r2}^2 + 4.\, \varepsilon_{r2} + \varepsilon_{c2}^2 + 4. - 2.\, LiquidFraction\, \varepsilon_{r2}^2 \\
& - 4.\, LiquidFraction\, \varepsilon_{r2} - 2.\, LiquidFraction\, \varepsilon_{c2}^2 + 4.\, \varepsilon_{r1}\, LiquidFraction \\
& + 2.\, \varepsilon_{r1}\, LiquidFraction\, \varepsilon_{r2}\big)
\end{aligned} \tag{1.16}$$

$\xrightarrow{\text{to string}}$

"(4.*LiquidFraction^2*epsilon[r2]*epsilon[r1]-2.*epsilon[r1]^2*
LiquidFraction^2-4.*epsilon[c2]*LiquidFraction*epsilon[c1]C epsilon[r1]* **(1.17)**

epsilon[c2]^2-4.*epsilon[r1]*LiquidFraction*epsilon[r2]C 4.*epsilon[c2]*LiquidFraction^2*epsilon[c1]-2.*epsilon[c1]^2*LiquidFraction^2C 2.*epsilon[r1]^2*LiquidFractionC 2.*LiquidFraction*epsilon[r2]^2-2.*epsilon[c2]^2*LiquidFraction^2C 4.*epsilon[r1]C 4.*epsilon[r1]*epsilon[r2]-4.*epsilon[r1]*LiquidFraction-2.*LiquidFraction^2*epsilon[r2]^2C epsilon[r1]*epsilon[r2]^2C 2.*LiquidFraction*epsilon[c2]^2C epsilon[r1]^2*LiquidFraction*epsilon[r2]-1.*epsilon[r1]*LiquidFraction*epsilon[r2]^2C 4.*LiquidFraction*epsilon[r2]-1.*epsilon[r1]*LiquidFraction*epsilon[c2]^2C epsilon[c1]^2*LiquidFraction*epsilon[r2]C 2.*epsilon[c1]^2*LiquidFraction)/(LiquidFraction^2*epsilon[r2]^2C epsilon[r1]^2*LiquidFraction^2-2.*LiquidFraction^2*epsilon[r2]*epsilon[r1]C epsilon[c2]^2*LiquidFraction^2C epsilon[c1]^2*LiquidFraction^2C 2.*epsilon[c2]*LiquidFraction*epsilon[c1]-2.*epsilon[c2]*LiquidFraction^2*epsilon[c1]C epsilon[r2]^2C 4.*epsilon[r2]C epsilon[c2]^2C 4.-2.*LiquidFraction*epsilon[r2]^2-4.*LiquidFraction*epsilon[r2]-2.*LiquidFraction*epsilon[c2]^2C 4.*epsilon[r1]*LiquidFractionC 2.*epsilon[r1]*LiquidFraction*epsilon[r2])"

$$epsilonEffC := \left(\varepsilon_{c1}\varepsilon_{r2}^2 - 4.\,\varepsilon_{c1}\,LiquidFraction + 4.\,\varepsilon_{c2}\,LiquidFraction + \varepsilon_{c1}\varepsilon_{c2}^2 + 4.\,\varepsilon_{c1}\varepsilon_{r2} - 1.\,\varepsilon_{c1}\,LiquidFraction\,\varepsilon_{r2}^2 + \varepsilon_{c2}\,LiquidFraction\,\varepsilon_{c1}^2 + 4.\,\varepsilon_{c2}\,LiquidFraction\,\varepsilon_{r1} + \varepsilon_{c2}\,LiquidFraction\,\varepsilon_{r1}^2 - 1.\,\varepsilon_{c1}\,LiquidFraction\,\varepsilon_{c2}^2 - 4.\,\varepsilon_{c1}\,LiquidFraction\,\varepsilon_{r2} + 4.\,\varepsilon_{c1}\right) \Big/ \left(LiquidFraction^2\,\varepsilon_{r2}^2 + \varepsilon_{r1}^2\,LiquidFraction^2 - 2.\,LiquidFraction^2\,\varepsilon_{r2}\varepsilon_{r1} + \varepsilon_{c2}^2\,LiquidFraction^2 + \varepsilon_{c1}^2\,LiquidFraction^2 + 2.\,\varepsilon_{c2}\,LiquidFraction\,\varepsilon_{c1} - 2.\,\varepsilon_{c2}\,LiquidFraction^2\,\varepsilon_{c1} + \varepsilon_{r2}^2 + 4.\,\varepsilon_{r2} + \varepsilon_{c2}^2 + 4. - 2.\,LiquidFraction\,\varepsilon_{r2}^2 - 4.\,LiquidFraction\,\varepsilon_{r2} - 2.\,LiquidFraction\,\varepsilon_{c2}^2 + 4.\,\varepsilon_{r1}\,LiquidFraction + 2.\,\varepsilon_{r1}\,LiquidFraction\,\varepsilon_{r2}\right)$$

$$\left(\varepsilon_{c1}\varepsilon_{r2}^2 - 4.\,\varepsilon_{c1}\,LiquidFraction + 4.\,\varepsilon_{c2}\,LiquidFraction + \varepsilon_{c1}\varepsilon_{c2}^2 + 4.\,\varepsilon_{c1}\varepsilon_{r2} - 1.\,\varepsilon_{c1}\,LiquidFraction\,\varepsilon_{r2}^2 + \varepsilon_{c2}\,LiquidFraction\,\varepsilon_{c1}^2 + 4.\,\varepsilon_{c2}\,LiquidFraction\,\varepsilon_{r1} + \varepsilon_{c2}\,LiquidFraction\,\varepsilon_{r1}^2 - 1.\,\varepsilon_{c1}\,LiquidFraction\,\varepsilon_{c2}^2 - 4.\,\varepsilon_{c1}\,LiquidFraction\,\varepsilon_{r2} + 4.\,\varepsilon_{c1}\right) \Big/ \left(LiquidFraction^2\,\varepsilon_{r2}^2 + \varepsilon_{r1}^2\,LiquidFraction^2 - 2.\,LiquidFraction^2\,\varepsilon_{r2}\varepsilon_{r1} + \varepsilon_{c2}^2\,LiquidFraction^2 + \varepsilon_{c1}^2\,LiquidFraction^2 + 2.\,\varepsilon_{c2}\,LiquidFraction\,\varepsilon_{c1} - 2.\,\varepsilon_{c2}\,LiquidFraction^2\,\varepsilon_{c1} + \varepsilon_{r2}^2 + 4.\,\varepsilon_{r2} + \varepsilon_{c2}^2 + 4. - 2.\,LiquidFraction\,\varepsilon_{r2}^2 - 4.\,LiquidFraction\,\varepsilon_{r2} - 2.\,LiquidFraction\,\varepsilon_{c2}^2 + 4.\,\varepsilon_{r1}\,LiquidFraction + 2.\,\varepsilon_{r1}\,LiquidFraction\,\varepsilon_{r2}\right) \tag{1.18}$$

$\xrightarrow{\text{to string}}$

```
"(epsilon[c1]*epsilon[r2]^2-4.*epsilon[c1]*LiquidFractionC 4.*epsilon[c2]*
LiquidFractionC epsilon[c1]*epsilon[c2]^2C 4.*epsilon[c1]*epsilon[r2]-1.*
epsilon[c1]*LiquidFraction*epsilon[r2]^2C epsilon[c2]*LiquidFraction*
epsilon[c1]^2C 4.*epsilon[c2]*LiquidFraction*epsilon[r1]C epsilon[c2]*
LiquidFraction*epsilon[r1]^2-1.*epsilon[c1]*LiquidFraction*epsilon[c2]^2
-4.*epsilon[c1]*LiquidFraction*epsilon[r2]C 4.*epsilon[c1])/
(LiquidFraction^2*epsilon[r2]^2C epsilon[r1]^2*LiquidFraction^2-2.*
LiquidFraction^2*epsilon[r2]*epsilon[r1]C epsilon[c2]^2*
LiquidFraction^2C epsilon[c1]^2*LiquidFraction^2C 2.*epsilon[c2]*
LiquidFraction*epsilon[c1]-2.*epsilon[c2]*LiquidFraction^2*epsilon[c1]C
epsilon[r2]^2C 4.*epsilon[r2]C epsilon[c2]^2C 4.-2.*LiquidFraction*epsilon
[r2]^2-4.*LiquidFraction*epsilon[r2]-2.*LiquidFraction*epsilon[c2]^2C 4.*
epsilon[r1]*LiquidFractionC 2.*epsilon[r1]*LiquidFraction*epsilon[r2])"
```
(1.19)

$optimize(epsilonEffR)$

$$t1 = LiquidFraction^2,\ t3 = t1\,\varepsilon_{r2}\varepsilon_{r1},\ t5 = \varepsilon_{r1}^2,\ t6 = t5\,t1,\ t9 = \varepsilon_{c2}\,LiquidFraction\,\varepsilon_{c1},\ t11 = \varepsilon_{c2}^2,\ t13 = \varepsilon_{r1}\,LiquidFraction,\ t14 = t13\,\varepsilon_{r2},\ t17 = \varepsilon_{c2}\,t1\,\varepsilon_{c1},\ t19 = \varepsilon_{c1}^2,\ t20 = t19\,t1,\ t22 = t5\,LiquidFraction,\ t24 = \varepsilon_{r2}^2,\ t26 = 2.\,LiquidFraction\,t24,\ t27 = t11\,t1,\ t30 = 4.\,t3 - 2.\,t6 - 4.\,t9 + \varepsilon_{r1}\,t11 - 4.\,t14 + 4.\,t17 - 2.\,t20 + 2.\,t22 + t26 - 2.\,t27 + 4.\,\varepsilon_{r1},\ t33 = 4.\,t13,\ t34 = t1\,t24,\ t38 = 2.\,LiquidFraction\,t11,\ t43 = 4.\,LiquidFraction\,\varepsilon_{r2},\ t46 = t19\,LiquidFraction,\ t49 = 4.\,\varepsilon_{r1}\varepsilon_{r2} - t33 - 2.\,t34 + \varepsilon_{r1}\,t24 + t38 + t22\,\varepsilon_{r2} - 1.\,t13\,t24 + t43 - 1.\,t13\,t11 + t46\,\varepsilon_{r2} + 2.\,t46,\ t56 = t34 + t6 - 2.\,t3 + t27 + t20 + 2.\,t9 - 2.\,t17 + t24 + 4.\,\varepsilon_{r2} + t11 + 4. - t26 - t43 - t38 + t33 + 2.\,t14,\ t58 = \frac{t30 + t49}{t56} \tag{1.20}$$

$optimize(epsilonEffC)$

$$t1 = \varepsilon_{r2}^2,\ t3 = \varepsilon_{c1}\,LiquidFraction,\ t5 = \varepsilon_{c2}\,LiquidFraction,\ t7 = \varepsilon_{c2}^2,\ t13 = \varepsilon_{c1}^2,\ t17 = \varepsilon_{r1}^2,\ t24 = \varepsilon_{c1}\,t1 - 4.\,t3 + 4.\,t5 + \varepsilon_{c1}\,t7 + 4.\,\varepsilon_{c1}\varepsilon_{r2} - 1.\,t3\,t1 + t5\,t13 + 4.\,t5\,\varepsilon_{r1} + t5\,t17 - 1.\,t3\,t7 - 4.\,t3\,\varepsilon_{r2} + 4.\,\varepsilon_{c1},\ t25 = LiquidFraction^2,\ t45 = \varepsilon_{r1}\,LiquidFraction,\ t49 = t25\,t1 + t17\,t25 - 2.\,t25\,\varepsilon_{r2}\varepsilon_{r1} + t7\,t25 + t13\,t25 + 2.\,t5\,\varepsilon_{c1} - 2.\,\varepsilon_{c2}\,t25\,\varepsilon_{c1} + t1 + 4.\,\varepsilon_{r2} + t7 + 4. - 2.\,LiquidFraction\,t1 - 4.\,LiquidFraction\,\varepsilon_{r2} - 2.\,LiquidFraction\,t7 + 4.\,t45 + 2.\,t45\,\varepsilon_{r2},\ t51 = \frac{t24}{t49} \tag{1.21}$$

Solution with 3 media

restart

$$eq1 := \frac{(\varepsilon_{eff}-1)}{\varepsilon_{eff}+2} = \frac{(\varepsilon_1-1)}{\varepsilon_1+2}\cdot\eta_1 + \frac{(\varepsilon_2-1)}{\varepsilon_2+2}\cdot\eta_2 + \frac{(\varepsilon_3-1)}{\varepsilon_3+2}\cdot\eta_3$$

$$\frac{\varepsilon_{eff}-1}{\varepsilon_{eff}+2} = \frac{(\varepsilon_1-1)\,\eta_1}{\varepsilon_1+2} + \frac{(\varepsilon_2-1)\,\eta_2}{\varepsilon_2+2} + \frac{(\varepsilon_3-1)\,\eta_3}{\varepsilon_3+2} \qquad \textbf{(2.1)}$$

$\varepsilon_{eff} := \varepsilon_{reff} + I\cdot\varepsilon_{ceff};\ \varepsilon_1 := \varepsilon_{r1} + I\cdot\varepsilon_{c1};\ \varepsilon_2 := \varepsilon_{r2} + I\cdot\varepsilon_{c2};\ \varepsilon_3 := \varepsilon_{r3} + I\cdot\varepsilon_{c3};$

$$\varepsilon_{reff} + \mathrm{I}\,\varepsilon_{ceff}$$

$$\varepsilon_{r1} + \mathrm{I}\,\varepsilon_{c1}$$

$$\varepsilon_{r2} + \mathrm{I}\,\varepsilon_{c2}$$

$$\varepsilon_{r3} + \mathrm{I}\,\varepsilon_{c3} \qquad \textbf{(2.2)}$$

$eq2 := evalc(eq1)$

$$\frac{(\varepsilon_{reff}-1)\,(\varepsilon_{reff}+2)}{(\varepsilon_{reff}+2)^2+\varepsilon_{ceff}^2} + \frac{\varepsilon_{ceff}^2}{(\varepsilon_{reff}+2)^2+\varepsilon_{ceff}^2} + \mathrm{I}\left(\frac{\varepsilon_{ceff}\,(\varepsilon_{reff}+2)}{(\varepsilon_{reff}+2)^2+\varepsilon_{ceff}^2} - \frac{(\varepsilon_{reff}-1)\,\varepsilon_{ceff}}{(\varepsilon_{reff}+2)^2+\varepsilon_{ceff}^2}\right) = \left(\frac{(\varepsilon_{r1}-1)\,(\varepsilon_{r1}+2)}{(\varepsilon_{r1}+2)^2+\varepsilon_{c1}^2} + \frac{\varepsilon_{c1}^2}{(\varepsilon_{r1}+2)^2+\varepsilon_{c1}^2}\right)\eta_1 + \left(\frac{(\varepsilon_{r2}-1)\,(\varepsilon_{r2}+2)}{(\varepsilon_{r2}+2)^2+\varepsilon_{c2}^2} + \frac{\varepsilon_{c2}^2}{(\varepsilon_{r2}+2)^2+\varepsilon_{c2}^2}\right)\eta_2 + \left(\frac{(\varepsilon_{r3}-1)\,(\varepsilon_{r3}+2)}{(\varepsilon_{r3}+2)^2+\varepsilon_{c3}^2} + \frac{\varepsilon_{c3}^2}{(\varepsilon_{r3}+2)^2+\varepsilon_{c3}^2}\right)\eta_3 + \mathrm{I}\left(\left(\frac{\varepsilon_{c1}\,(\varepsilon_{r1}+2)}{(\varepsilon_{r1}+2)^2+\varepsilon_{c1}^2} - \frac{(\varepsilon_{r1}-1)\,\varepsilon_{c1}}{(\varepsilon_{r1}+2)^2+\varepsilon_{c1}^2}\right)\eta_1 + \left(\frac{\varepsilon_{c2}\,(\varepsilon_{r2}+2)}{(\varepsilon_{r2}+2)^2+\varepsilon_{c2}^2} - \frac{(\varepsilon_{r2}-1)\,\varepsilon_{c2}}{(\varepsilon_{r2}+2)^2+\varepsilon_{c2}^2}\right)\eta_2 + \left(\frac{\varepsilon_{c3}\,(\varepsilon_{r3}+2)}{(\varepsilon_{r3}+2)^2+\varepsilon_{c3}^2} - \frac{(\varepsilon_{r3}-1)\,\varepsilon_{c3}}{(\varepsilon_{r3}+2)^2+\varepsilon_{c3}^2}\right)\eta_3\right) \qquad \textbf{(2.3)}$$

$$eq2r := \frac{(\varepsilon_{reff}-1)\,(\varepsilon_{reff}+2)}{(\varepsilon_{reff}+2)^2+\varepsilon_{ceff}^2} + \frac{\varepsilon_{ceff}^2}{(\varepsilon_{reff}+2)^2+\varepsilon_{ceff}^2} = \left(\frac{(\varepsilon_{r1}-1)\,(\varepsilon_{r1}+2)}{(\varepsilon_{r1}+2)^2+\varepsilon_{c1}^2} + \frac{\varepsilon_{c1}^2}{(\varepsilon_{r1}+2)^2+\varepsilon_{c1}^2}\right)\eta_1 + \left(\frac{(\varepsilon_{r2}-1)\,(\varepsilon_{r2}+2)}{(\varepsilon_{r2}+2)^2+\varepsilon_{c2}^2} + \frac{\varepsilon_{c2}^2}{(\varepsilon_{r2}+2)^2+\varepsilon_{c2}^2}\right)\eta_2$$

$$+\left(\frac{(\varepsilon_{r3}-1)\,(\varepsilon_{r3}+2)}{(\varepsilon_{r3}+2)^2+\varepsilon_{c3}^2}+\frac{\varepsilon_{c3}^2}{(\varepsilon_{r3}+2)^2+\varepsilon_{c3}^2}\right)\eta_3$$

$$\frac{(\varepsilon_{reff}-1)\,(\varepsilon_{reff}+2)}{(\varepsilon_{reff}+2)^2+\varepsilon_{ceff}^2}+\frac{\varepsilon_{ceff}^2}{(\varepsilon_{reff}+2)^2+\varepsilon_{ceff}^2}=\left(\frac{(\varepsilon_{r1}-1)\,(\varepsilon_{r1}+2)}{(\varepsilon_{r1}+2)^2+\varepsilon_{c1}^2}+\frac{\varepsilon_{c1}^2}{(\varepsilon_{r1}+2)^2+\varepsilon_{c1}^2}\right)\eta_1+\left(\frac{(\varepsilon_{r2}-1)\,(\varepsilon_{r2}+2)}{(\varepsilon_{r2}+2)^2+\varepsilon_{c2}^2}+\frac{\varepsilon_{c2}^2}{(\varepsilon_{r2}+2)^2+\varepsilon_{c2}^2}\right)\eta_2+\left(\frac{(\varepsilon_{r3}-1)\,(\varepsilon_{r3}+2)}{(\varepsilon_{r3}+2)^2+\varepsilon_{c3}^2}+\frac{\varepsilon_{c3}^2}{(\varepsilon_{r3}+2)^2+\varepsilon_{c3}^2}\right)\eta_3 \quad \textbf{(2.4)}$$

$$eq2c:=\left(\frac{\varepsilon_{ceff}\,(\varepsilon_{reff}+2)}{(\varepsilon_{reff}+2)^2+\varepsilon_{ceff}^2}-\frac{(\varepsilon_{reff}-1)\,\varepsilon_{ceff}}{(\varepsilon_{reff}+2)^2+\varepsilon_{ceff}^2}\right)=\left(\left(\frac{\varepsilon_{c1}\,(\varepsilon_{r1}+2)}{(\varepsilon_{r1}+2)^2+\varepsilon_{c1}^2}-\frac{(\varepsilon_{r1}-1)\,\varepsilon_{c1}}{(\varepsilon_{r1}+2)^2+\varepsilon_{c1}^2}\right)\eta_1+\left(\frac{\varepsilon_{c2}\,(\varepsilon_{r2}+2)}{(\varepsilon_{r2}+2)^2+\varepsilon_{c2}^2}-\frac{(\varepsilon_{r2}-1)\,\varepsilon_{c2}}{(\varepsilon_{r2}+2)^2+\varepsilon_{c2}^2}\right)\eta_2+\left(\frac{\varepsilon_{c3}\,(\varepsilon_{r3}+2)}{(\varepsilon_{r3}+2)^2+\varepsilon_{c3}^2}-\frac{(\varepsilon_{r3}-1)\,\varepsilon_{c3}}{(\varepsilon_{r3}+2)^2+\varepsilon_{c3}^2}\right)\eta_3\right)$$

$$\frac{\varepsilon_{ceff}\,(\varepsilon_{reff}+2)}{(\varepsilon_{reff}+2)^2+\varepsilon_{ceff}^2}-\frac{(\varepsilon_{reff}-1)\,\varepsilon_{ceff}}{(\varepsilon_{reff}+2)^2+\varepsilon_{ceff}^2}=\left(\frac{\varepsilon_{c1}\,(\varepsilon_{r1}+2)}{(\varepsilon_{r1}+2)^2+\varepsilon_{c1}^2}-\frac{(\varepsilon_{r1}-1)\,\varepsilon_{c1}}{(\varepsilon_{r1}+2)^2+\varepsilon_{c1}^2}\right)\eta_1+\left(\frac{\varepsilon_{c2}\,(\varepsilon_{r2}+2)}{(\varepsilon_{r2}+2)^2+\varepsilon_{c2}^2}-\frac{(\varepsilon_{r2}-1)\,\varepsilon_{c2}}{(\varepsilon_{r2}+2)^2+\varepsilon_{c2}^2}\right)\eta_2+\left(\frac{\varepsilon_{c3}\,(\varepsilon_{r3}+2)}{(\varepsilon_{r3}+2)^2+\varepsilon_{c3}^2}-\frac{(\varepsilon_{r3}-1)\,\varepsilon_{c3}}{(\varepsilon_{r3}+2)^2+\varepsilon_{c3}^2}\right)\eta_3 \quad \textbf{(2.5)}$$

$sol := solve(\{eq2r, eq2c\}, \{\varepsilon_{reff}, \varepsilon_{ceff}\})$
$eps_c := rhs(sol[1]):$
$eps_r := rhs(sol[2]):$

Numerical application

$\varepsilon_1 := -9.04639807977 + I\cdot 28.8437268491;\ \eta_1 := (1e0-\eta_2-\eta_3);$

$$-9.04639807977+28.8437268491\,\mathrm{I}$$

$$0.7308 \quad \textbf{(2.6)}$$

$\varepsilon_2 := -0.62993 + I\cdot 24.915;\ \eta_2 := 16.87e-2;\#Cr$

$$-0.62993+24.915\,\mathrm{I}$$

$$0.1687 \tag{2.7}$$

$\varepsilon_3 := -17.8077402901\mathrm{e}0 + I \cdot 28.97412511\mathrm{e}0;\ \eta_3 := 10.05\mathrm{e}{-2};$#*Ni*

$$-17.8077402901 + 28.97412511\,\mathrm{I}$$

$$0.1005 \tag{2.8}$$

$\varepsilon_{r1} := \mathrm{Re}(\varepsilon_1) : \varepsilon_{c1} := \mathrm{Im}(\varepsilon_1) : \varepsilon_{r2} := \mathrm{Re}(\varepsilon_2) : \varepsilon_{c2} := \mathrm{Im}(\varepsilon_2) : \varepsilon_{r3} := \mathrm{Re}(\varepsilon_3) : \varepsilon_{c3} := \mathrm{Im}(\varepsilon_3) :$

$eps_r;$

$$-7.977157246 \tag{2.9}$$

eps_c

$$28.75781330 \tag{2.10}$$

restart

Formula used by Inam Mirza.

$\varepsilon_{eff} := \varepsilon_m \cdot \left(1 + \frac{3 \cdot f1 \cdot \mathrm{beta}}{1 - f1 \cdot \mathrm{beta}}\right); \mathrm{beta} := \frac{(\varepsilon_{np} - \varepsilon_m)}{\varepsilon_{np} - 2 \cdot \varepsilon_m}$

$$1.625 + \frac{(3.475945863 + 1.205338949\,\mathrm{I})\ f1}{1 + (-0.7130145361 - 0.2472490151\,\mathrm{I})\ f1}$$

$$0.7130145361 + 0.2472490151\,\mathrm{I} \tag{1}$$

$\varepsilon_{np} := 0 - I \cdot 2.8;\ f1 :=$

$$-2.8\,\mathrm{I} \tag{2}$$

$\varepsilon_m := 1.625 - 0 \cdot I$

$$1.625 \tag{3}$$

$evalc(\varepsilon_{eff})$

$$1.625 + \frac{3.475945863\ f1\ (1 - 0.7130145361\ f1)}{(1 - 0.7130145361\ f1)^2 + 0.06113207547\ f1^2} - \frac{0.2980188680\ f1^2}{(1 - 0.7130145361\ f1)^2 + 0.06113207547\ f1^2} + \mathrm{I}\left(\frac{1.205338949\ f1\ (1 - 0.7130145361\ f1)}{(1 - 0.7130145361\ f1)^2 + 0.06113207547\ f1^2} + \frac{0.8594241912\ f1^2}{(1 - 0.7130145361\ f1)^2 + 0.06113207547\ f1^2}\right) \tag{4}$$